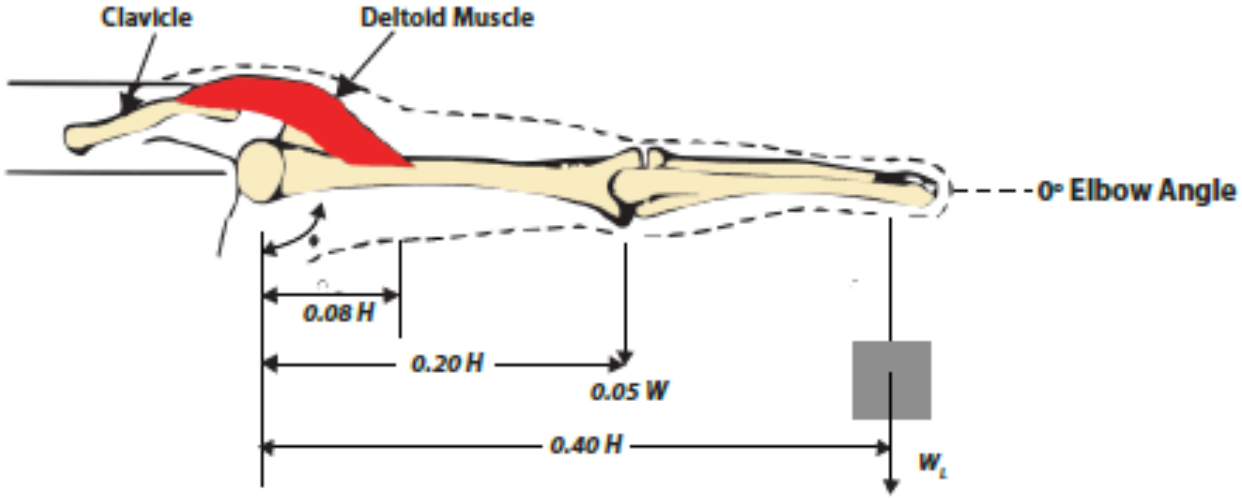
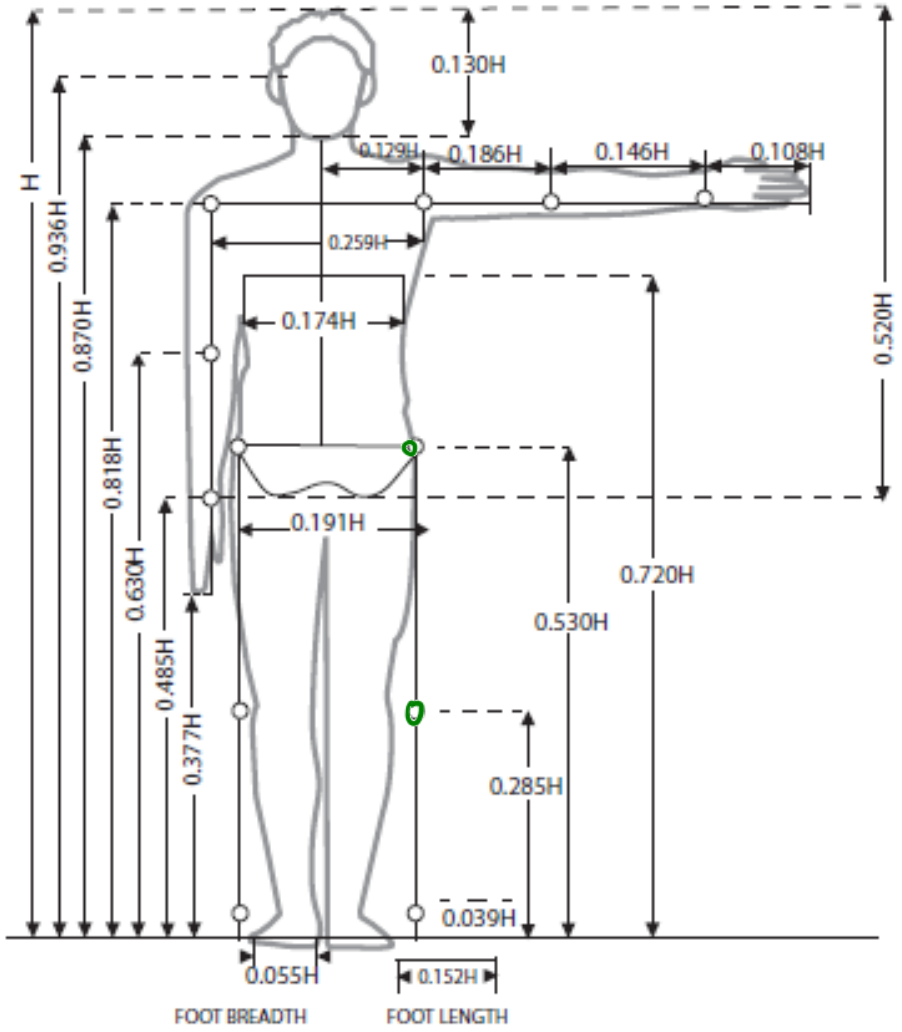
Static analysis - shoulder



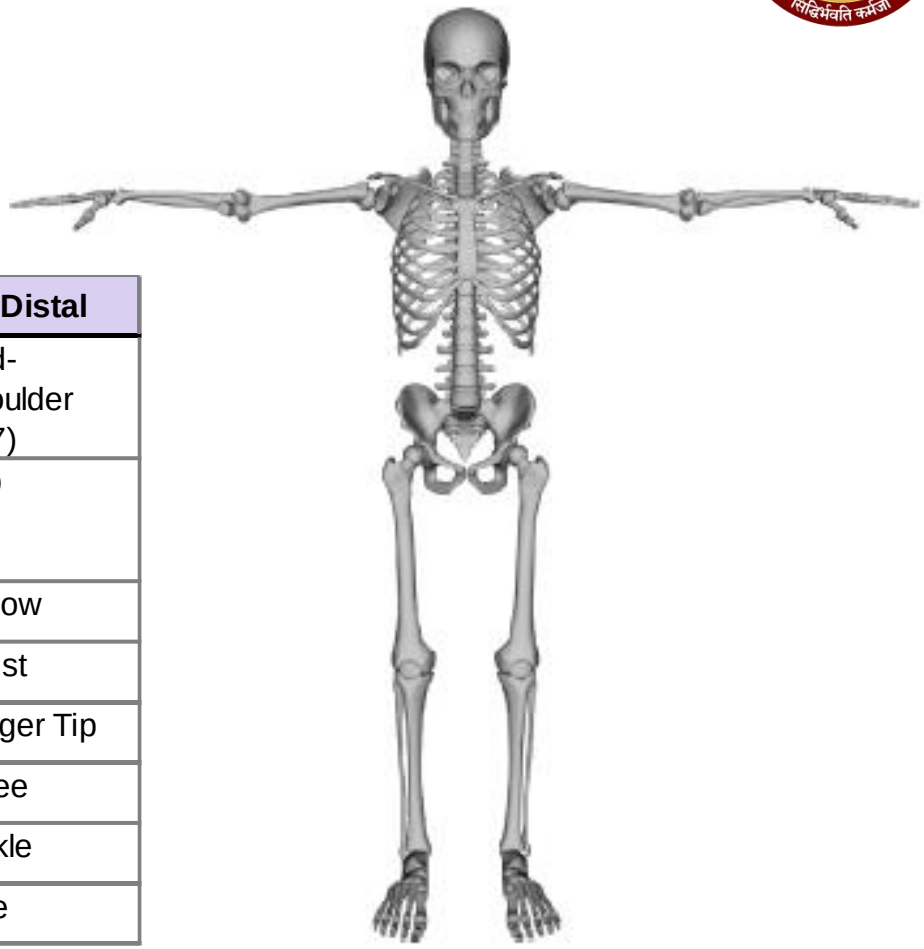


Anthropometric data

Anthropometry: scientific study of measurements of the human body.



Segment	Proximal	Distal
Head & Neck	Vertex	Mid-shoulder (C7)
Trunk	Mid-shoulder (C7)	Hip
Upper Arm	Shoulder	Elbow
Forearm	Elbow	Wrist
Hand	Wrist	Finger Tip
Thigh	Hip	Knee
Shank	Knee	Ankle
Foot	Heel	Toe



Anthropometric data referenced from IP Herman, Physics of the Human Body and DA Winter, Biomechanics and Motor Control of Human Movement

Anthropometric data

Segmental masses as % of BW

Segment	Males	Females	Average
Head & Neck	6.94	6.68	6.81
Trunk	43.46	42.58	43.02
Upper Arm	2.71	2.55	2.63
Forearm	1.62	1.38	1.5
Hand	0.61	0.56	0.585
Thigh	14.16	14.78	14.47
Shank	4.33	4.81	4.57
Foot	1.37	1.29	1.33

Segmental CG locations

% segment length (from proximal end)

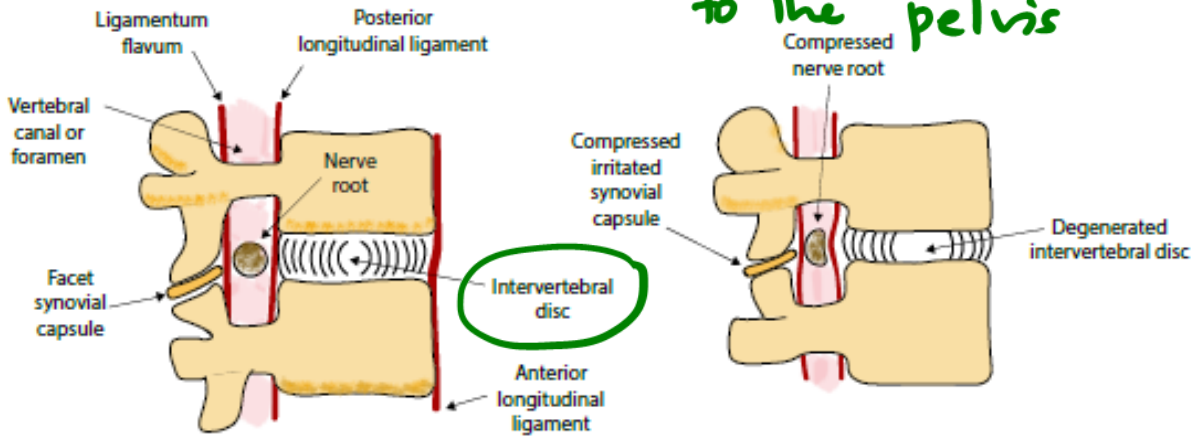
Segment	Males	Females	Average
Head & Neck	50.02	48.41	49.22
Trunk	43.1	37.82	40.46
Upper Arm	57.72	57.54	57.63
Forearm	45.74	45.59	45.67
Hand	79	74.74	76.87
Thigh	40.95	36.12	38.54
Shank	43.95	43.52	43.74
Foot	44.15	40.14	42.15

Segment	Proximal	Distal
Head & Neck	Vertex	Mid-shoulder (C7)
Trunk	Mid-shoulder (C7)	Hip
Upper Arm	Shoulder	Elbow
Forearm	Elbow	Wrist
Hand	Wrist	Finger Tip
Thigh	Hip	Knee
Shank	Knee	Ankle
Foot	Heel	Toe

Paolo de Leva (1996) Adjustments to Zatsiorsky-Seluyanov's Segment Inertia Parameters . Journal of Biomechanics 29 (9), pp. 1223-1230

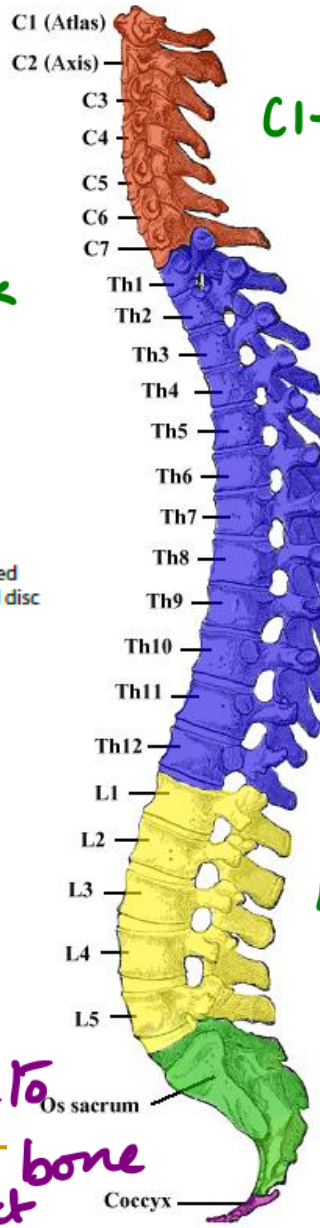
Statics - Spine

- Protects the spinal cord
- supports the head, neck & upper extremities
- transfers loads from head & trunk to the pelvis



- provides movement

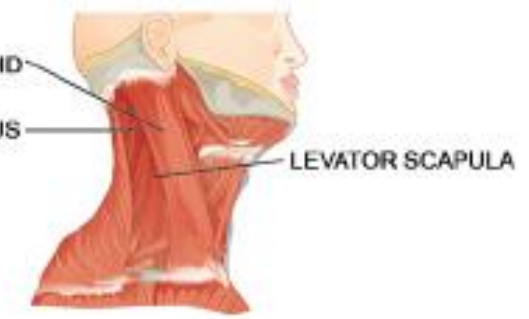
Intervertebral disk - fibrous cartilage shock absorbers, reduce bone-to-bone contact



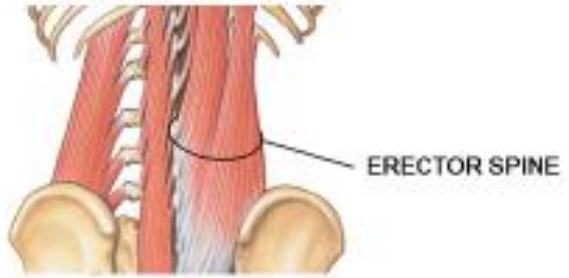
C1-C7

T1-T12

L1-L5



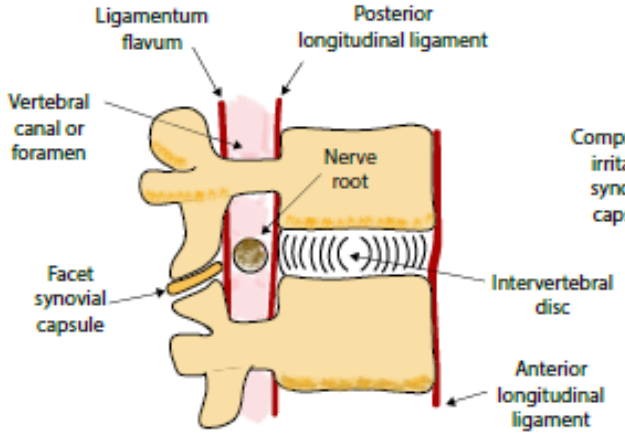


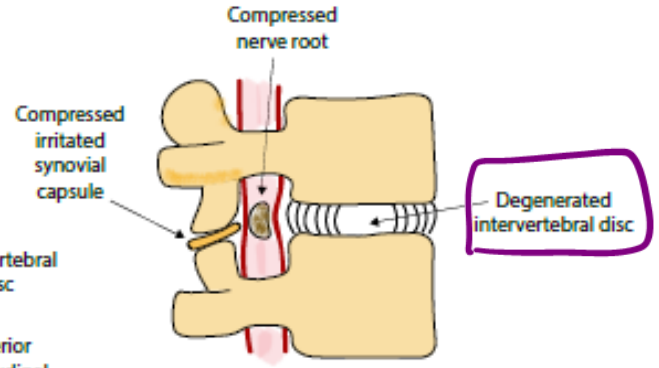


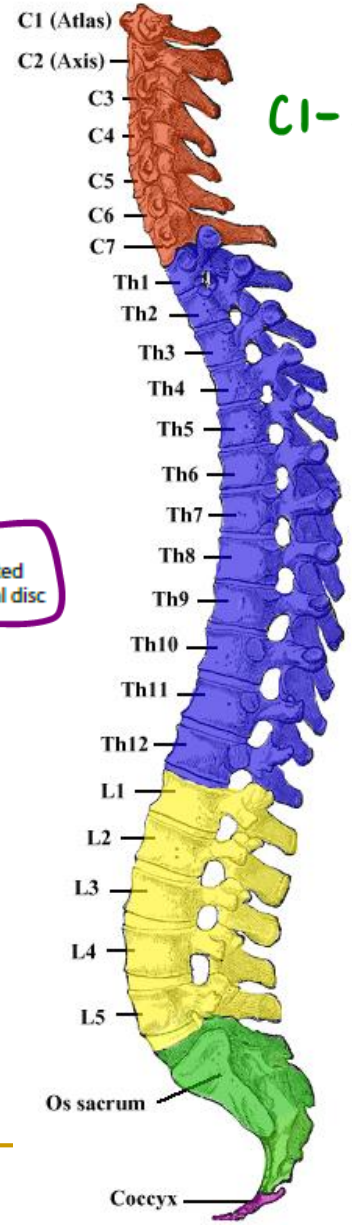
TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

Henry Vandyke Carter [Public domain or Public domain], via Wikimedia Commons

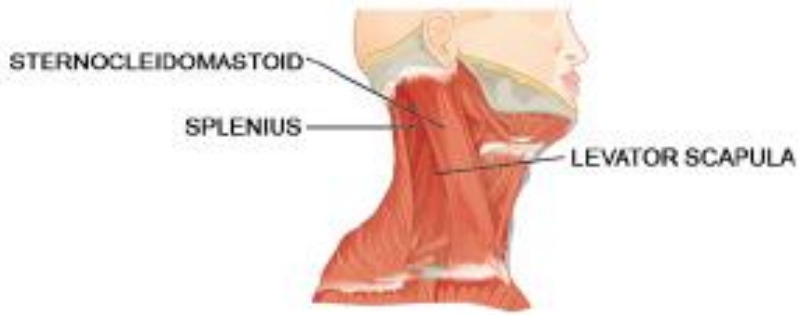
Statics - Spine







C1-C7





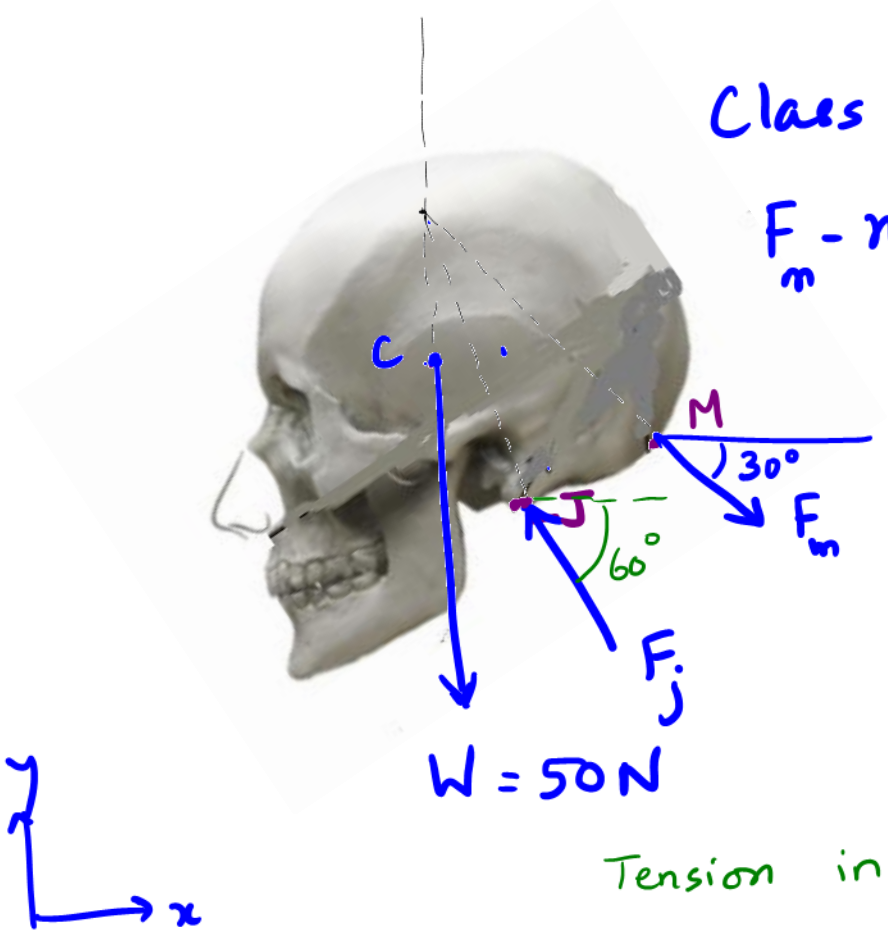


TTK Center for Rehabilitation Research & Device Development (R2D2)
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Static analysis – spine

Atlanto occipital joint - double condyloid jt. between Atlas C1 & occipital bone of the head



Class 1 lever

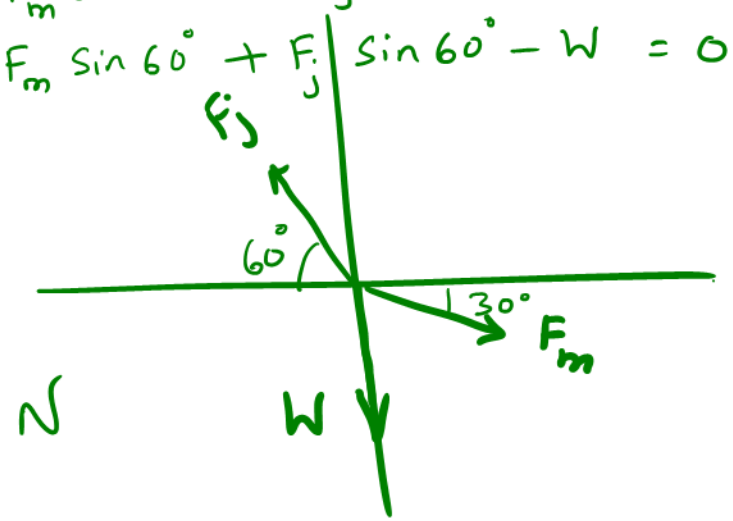
F_m - neck extensor

location, direction known

$\Sigma F_x = 0 \rightarrow F_m \cos 30^\circ - F_j \cos 60^\circ = 0$

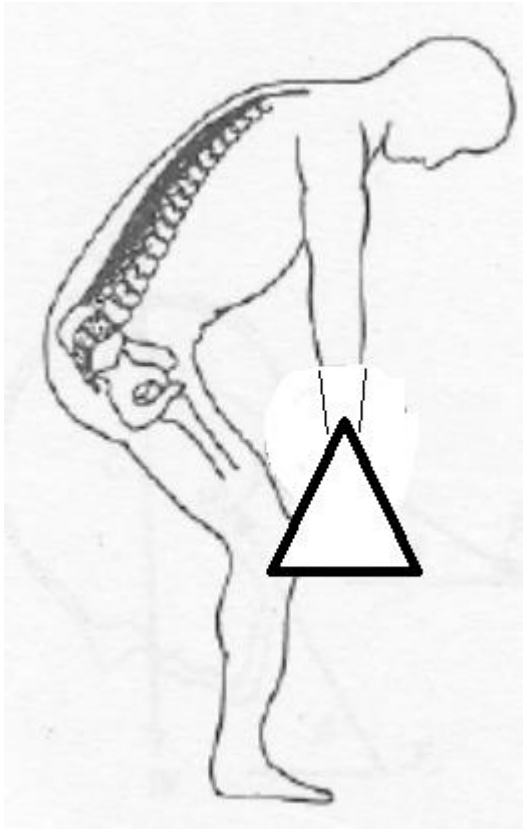
$\Sigma F_y = 0 \rightarrow -F_m \sin 60^\circ + F_j \sin 60^\circ - W = 0$

$\Sigma M_c = 0$



Tension in neck extensor = 87 N

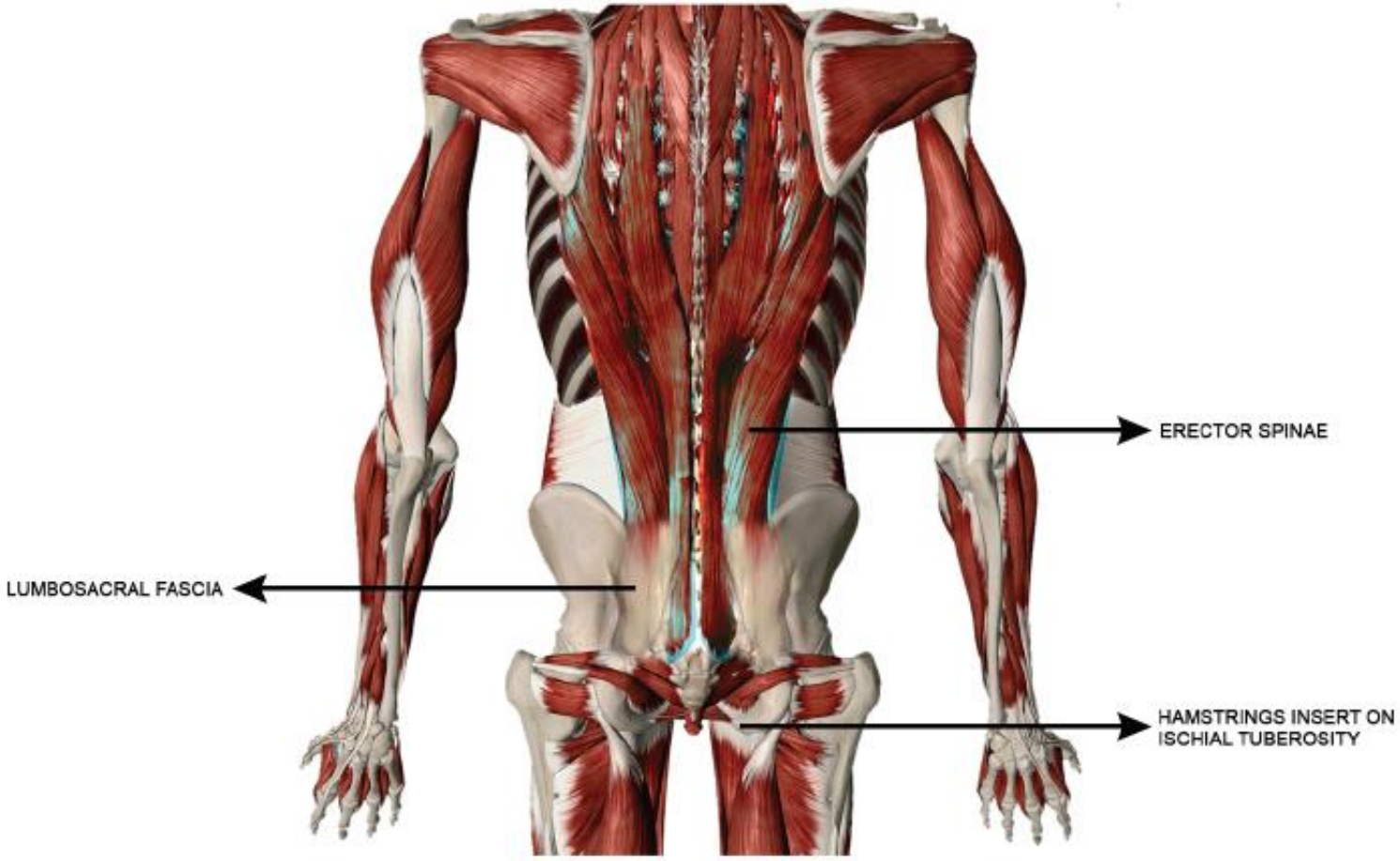
Static analysis – spine

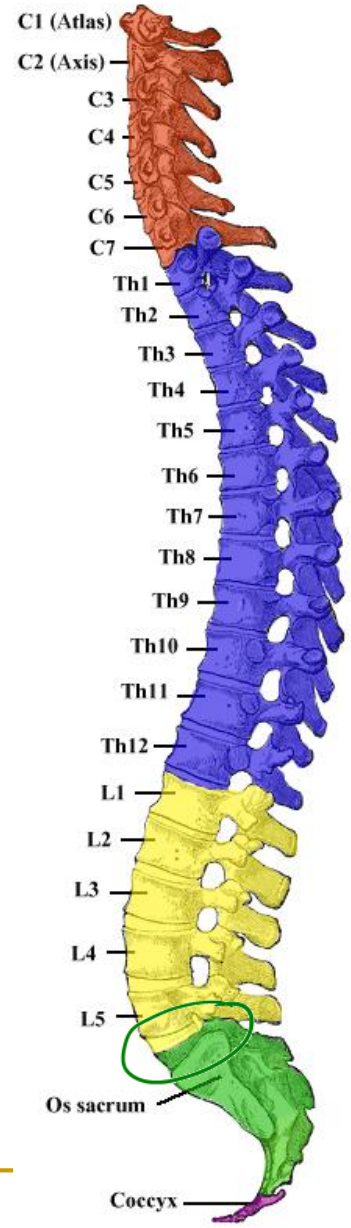






Static analysis – spine

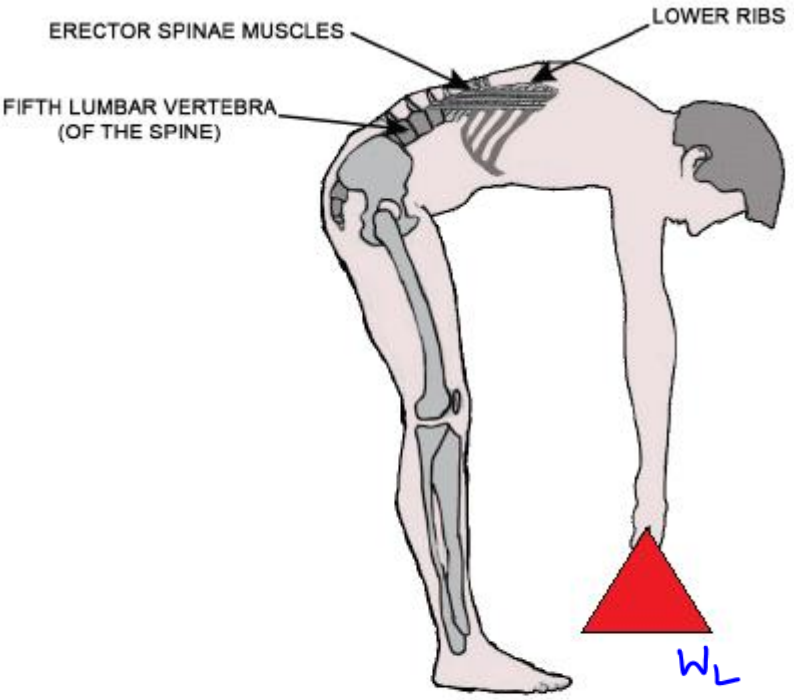




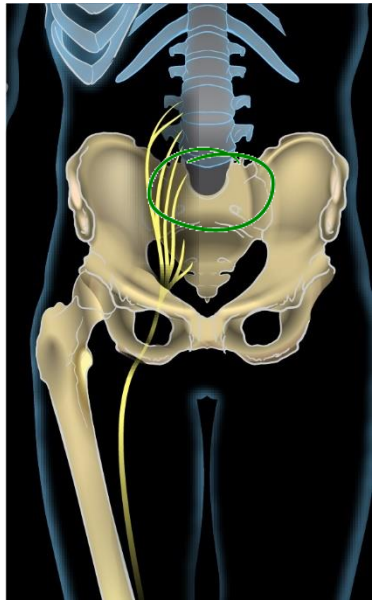
Lumbosacral joint

Henry Vandyke Carter [Public domain or Public domain], via Wikimedia Commons

Static analysis – spine

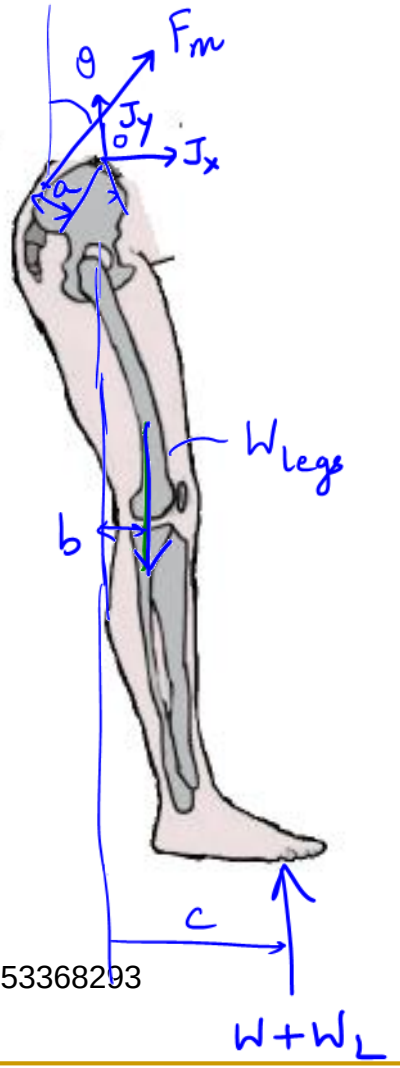


weight of person is W



By KDS4444 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=53368293

Why you shouldn't bend & lift



$\theta \rightarrow$ trunk flexion

Assume F_m acts parallel to the trunk

Let $\theta = 45^\circ$

$a = 0.02h$

$b = 0.08h$

$c = 0.12h$

$W_{legs} = 0.4W$

W_L

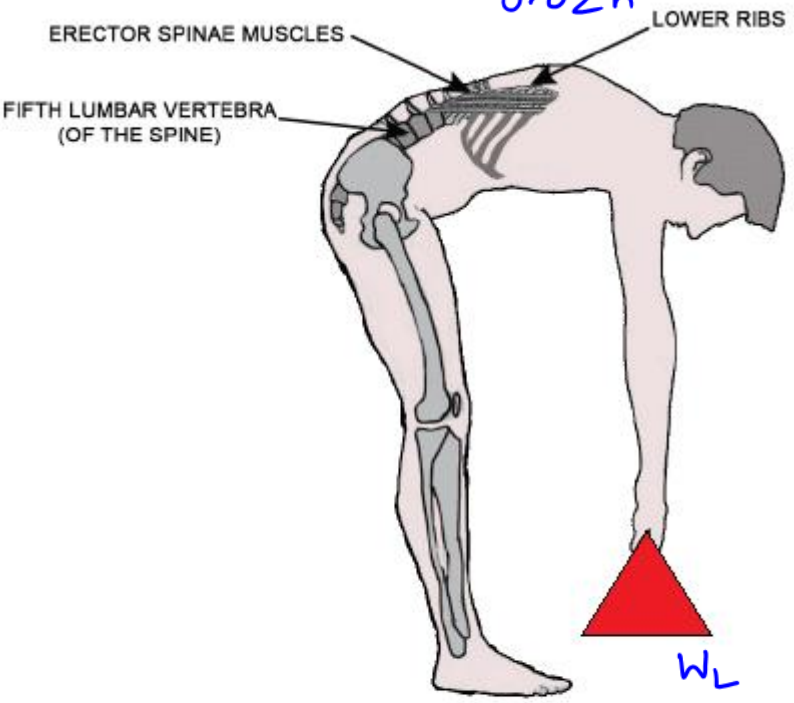
Static analysis – spine

Why you shouldn't bend & lift

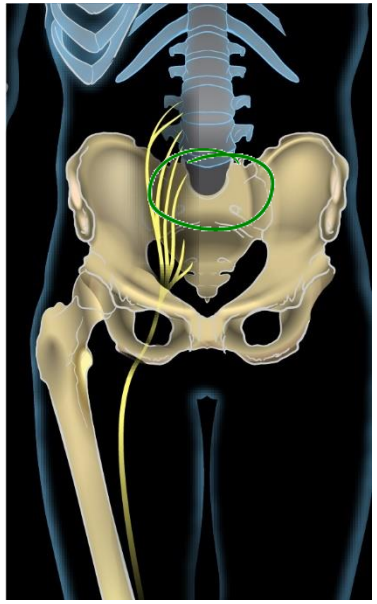
$\Sigma M_0 = 0$

$F_m = \dfrac{(0.12h)(W+W_L) - 0.08h(0.4W)}{0.02h}$

$= 4.4W + 6W_L$



weight of person is W



By KDS4444 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=53368293

θ_f

$W_L = 0.5W$

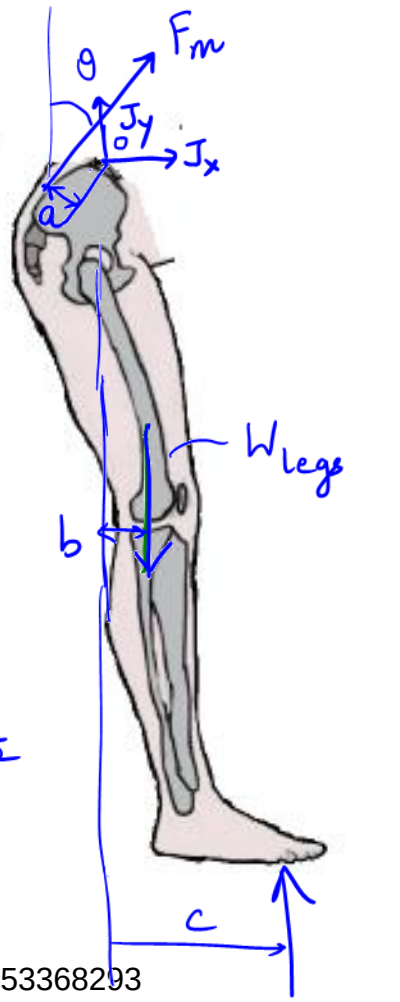
$F_m = 7.4W$

θ_f

$W_L = W$

$J = \sqrt{J_x^2 + J_y^2}$

$= 11.7W$



$\theta \rightarrow$ trunk flexion

Assume F_m acts parallel to the trunk

Let $\theta = 45^\circ$

$a = 0.02h$

$b = 0.08h$

$c = 0.12h$

$W_{legs} = 0.4W$

W_L

for $W_L = 0.5W$

$J = 8.2W$

$J_x = -F_m \sin\theta$

$= 5.2W$

$J_y = -(W+W_L) + W_{legs} - F_m \cos\theta = 6.3W$